

Anki deck i(patty)i made of all the questions:

https://drive.google.com/file/d/12Jc62nkgDs7__8EDPBxclCi91-KB8VMh/view?usp=sharing

Please let me know if there are any errors

Review strain patterns: <https://www.youtube.com/watch?v=8VYRhXALz08>

<https://youtu.be/PsqmpvHyZDw?si=6ZCyJFDDxr12d32n>

Week 7: Direct Study: Headache COAR - Wolf

Clinical Skills

1. Identify important aspects of a HPI relevant to the patient who presents with a headache

- Age
- Gender
- Aura
- Visual changes
- Paresthesia
- N/V (nausea and vomiting)
- Photophobia
- Phonophobia
- Tinnitus
- Traveling
- Trauma
- Smoking
- STIs/STDs
- Life stressors
- Meds: OCPs (contraceptives)
- Meds: OTC analgesic
- Worst HA of life
- Whiplash
- Pain: Uni or bi
- Fam hx
- Autonomic signs
- Syncope
- LOC
- Fever
- Mental status
- Rash
- Vital signs
- Illicit drug use
- Toxin exposure

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Pages 38 - 48

2. Perform an integrated physical exam of a patient with a headache, including HEENT, Neurological, cardiovascular and musculoskeletal considerations

Anterior Seated

Head and Neck

- ☐ Observe: General posture (including shoulders) and head carriage, facial symmetry
- ☐ Inspect: scalp, skull, hair for symmetry, trauma or lesions, AROM of cervical spine
- ☐ Palpate: face, suboccipital muscles and neck for muscle hypertonicity, lymphadenopathy, trachea Chapman's points, sinus pressure tenderness, palpation of supraorbital, infraorbital and mental foramen
- ☐ TMJ palpation, jaw opening/closing, palpation of temporal region and mastoid
- ☐ CN Exam: Facial sensation/jaw movements/corneal reflexes (CN V), Facial symmetry and strength (CN VII)

Eyes

- ☐ Observe: symmetry, pupil size, erythema
- ☐ Inspect: conjunctive and sclera
- ☐ Pupillary reaction and accommodation (Direct and consensual) (CN II/III)
- ☐ Fundoscopic Exam for papilledema
- ☐ Visual acuity and visual field testing (CN II)
- ☐ CN Exam: EOM (CN III, IV, VI), check for nystagmus

Ears

- ☐ Observe: ear position
- ☐ Inspect: External ear and external auditory canal
- ☐ Otoscope Exam: auditory canal and tympanic membranes
- ☐ Palpate: Mastoid process, temporal region
- ☐ Auditory acuity (whispered voice test)
- ☐ CN Exam: Hearing (CN VIII)

Nose

- ☐ Observe: symmetry, nasal septum, nasal mucosa
- ☐ Inspect: Nares with otoscope
- ☐ CN Exam: Smell (CN I)

Throat (Mouth and Pharynx)

- ☐ Observe: lips and muscular folds around the mouth, oral mucosa
- ☐ Inspect: tongue, palate, oropharynx, dentition, tonsils
- ☐ Palpate: temporomandibular joint for clicks or deviation (if not already done above)
- ☐ CN Exam: Bite (CN V), Tongue symmetry/position/movement (CN XII), Uvula deviation (CN X), swallowing and rise of the palate/gag reflex (CN IX, X)

Neuro

- ☐ Observe: gait (tandem walk, heel/toe), speech
- ☐ Inspect: muscle strength, muscle bulk, muscle tone, coordination, rapid alternating movement, sensation, proprioception, standing flexion test
- ☐ Deep Tendon Reflexes: upper and lower extremities
- ☐ Plantar Response (Babinski), Romberg Test, Pronator Drift, Hoffman reflex
- ☐ CN Exam: Voice and Speech (CN V, VII, X, XII), shoulder and neck movements (CN XI)

Posterior Seated

- ☐ Observe: position of the thorax and cervical spine, cervical AROM
- ☐ Palpate: Thyroid, swallow, cervical PROM, paraspinals, scalenes, SCM, upper trapezius, Zink Screen-O/A, C/T, T/L junctions, Thoracic spine (short lever) and TART
- ☐ Perform: Auditory acuity

Standing

- ☐ Observe gait (tandem walk, walking on heels, walking on toes) general posture
- ☐ Observe: standing flexion test if time permits
- ☐ Perform: Romberg test

Supine

- ☐ Observe: neck mobility, nuchal rigidity (Brudzinski sign, Kernig sign)
- ☐ Palpate: Cranial motion- vault hold, static symmetry, dynamic symmetry, SBS motion, temporal motion, parietal motion, Occipitomastoid suture, cervical anterior and posterior tenderpoints, sacral motion if time

3. Review the cranial nerve exam

4. Create and narrow down a differential diagnosis for headache

- Cerebral aneurysm

- Stroke/Cerebrovascular Accident (CVA) or transient ischemic attack (TIA)
- Dissection: internal carotid artery, vertebral artery
- Hemorrhage: epidural, subdural, or subarachnoid
- Chiari malformation
- Infections: meningitis, encephalitis (herpes simplex)
- Space occupying lesion (tumor, abscess)
- Pseudotumor cerebri
- Vasculitis, such as temporal arteritis
- Cluster headache
- Migraine Headache (Variants- Common, classic, hemiplegic, retinal, chronic)
- Occipital Neuralgia
- Paroxysmal hemicrania
- Tension Headache
- Nummular Headache
- TMJ dysfunction
- Sinusitis
- Tolosa-Hunt Syndrome
- Trigeminal Neuralgia
- Acute Glaucoma
- Post-concussion syndrome
- Psychiatric disease (somatization)
- **Rebound headache also known as medication overuse**
- Illicit drug use or prescription medication
- Whiplash injury, cervical strain
- Somatic dysfunction
- Cervicogenic headache
- Eye strain
- Optic neuritis
- Pheochromocytoma
- Carbon monoxide poisoning

5. Create an osteopathic treatment plan for the patient with a headache

- Make an accurate *clinical* diagnosis - primary vs secondary
- Make an accurate *structural* diagnosis
- Treating with OMT
 - SBS and temporal bone with OCMM
 - Treat cervical diagnosis - MFR, BLT, ME; careful not to over-stimulate the Vagus
 - Counterstrain
- Identify and avoid triggers

6. Describe pharmacologic therapy for the patient with headache

- **Abortive**
 - Selective 5-HT_{1D} or -1B agonists (**triptans**)
- **Prophylaxis**
 - Beta blockers, CCB (verapamil), TCAs, Anticonvulsants, Botox
- For **mild**
 - NSAIDs +/- antiemetics (dopamine antagonists = metoclopramide, prochlorperazine, chlorpromazine)
 - Triptans if NSAIDs fail
- For **mod-to-severe**
 - Triptans + NSAIDs
 - If no history of Substance Use Disorder = Opioid

7. Describe red flags related to headaches

- sudden onset
- worsening pattern (increasing frequency or severity of headaches)
- **Age greater than 50**
- new onset of headaches with an underlying medical condition (cancer, Lyme disease, HIV)
- headache with **concomitant systemic illness** (fever, neck stiffness, cutaneous rash)
- **focal neurologic signs or symptoms**
- **papilledema**
- headache **secondary to trauma**
- headache during pregnancy or postpartum
- prolonged or bizarre aura
- triggered by cough, exertion or Valsalva
- headache with trauma

Osteopathic Manipulative Medicine

1. Integrate an osteopathic structural exam into your physical exam

2. Discuss the A (Autonomics), B (Biomechanics), C (Circulation), and S (Screening) aspects of how structural findings could contribute to headache

- Autonomics
 - Sympathetics = **T1-4** → superior cervical ganglion (anterior to C2 transverse process) → travel along ICA, ECA, and CN 2, 3, 4, 6 & 9
 - Parasympathetics = **CN 3, 7, 9 & 10** → ciliary, sphenopalatine, submaxillary & otic ganglia
 - Be mindful of the jugular foramen, temporal bone, SBS, and sphenosquamous pivot
- Biomechanics
 - Head
 - Cervical spine
 - Thoracic spine
 - Sacrum/Pelvis
- Circulation
 - Tentorium cerebelli
 - Thoracic inlet
 - Diaphragm
- Screening
 - Postural - upper crossed syndrome
 - Leg length discrepancy or sacral base unleveling
 - Mind, body, spirit - increased cortisol and increased sympathetic tone

3. Perform osteopathic manipulative treatment aimed to assist a patient who presents with a headache

- **TMJD considerations:** Assessing for temporomandibular joint dysfunction (TMD) **should be done** in anyone with chronic headaches. Assessment should include:
 - Observing jaw opening and closing
 - palpation of **temporalis, buccinator, and medial/lateral pterygoids**
 - Cranial strain pattern at the SBS (remember the attachments of the pterygoids at the pterygoid plates of the sphenoid) and motion of the mandible and temporal bone (internal rotation or external rotation)
 - OA, AA and Upper Cervical Spine
 - Innervation of the TMJ is from the mandibular branch of the trigeminal nerve.
- **Occipitomastoid (OM) Suture Release** may assist with venous drainage from the cranium as well as assist in restoring motion to the *occiput* and *temporal* bone. Recall that CSF drains through arachnoid granulations into the venous sinuses. The venous sinuses ultimately drain from the sigmoid sinus into the internal jugular vein which passes through the jugular foramen. The jugular foramen is anteromedial to the OM suture.

- **OA BLT**



1. Slide one hand under the head to “palm the head”- middle finger towards OA, index and ring fingers align along the condyles
2. Place your other hand's index finger on the arch of the atlas (your patient may need to chin tuck a bit to find it)
3. Stabilize the atlas with your index finger- gently pull (flexion) or push (extension) and sideslip as necessary to find a balanced tension (not ease, not restriction but right in between). Hold until you feel a release & reassess.

4. Identify how viscerosomatic reflexes may contribute to headaches

Anatomy

1. Review innervation/ nociceptive patterns of the head and cranial dura and their anatomic relationships, including: Cervical plexus, Greater Occipital N., Lesser Occipital N., Trigeminal N., Vagus N.

- Ventral cervical plexus arises from **C1-4** → sensory component exits posterior to the SCM → sensory to head/neck
 - **Lesser occipital n. C2** - Sensory to skin of the **lateral head posterior to the ear**
 - **Great auricular n. C2,3** - Sensory to the **inferior ear, skin over the parotid gland and mastoid process**
 - **Transverse cervical n. C2,3** - Sensory to the **entirety of the anterior neck** (*passes deep to the platysma and sends sensory branches through the muscle.*)
 - **Supraclavicular n. C3,4** - Divides into medial, intermediate and lateral branches to supply sensory innervation over the **clavicle, pectoralis major, deltoid and parts of the posterior shoulder.**
 - **Dural branches - Nerves from the C2,3** spinal nerves will *ascend through the foramen magnum*, but also travel with the vagus nerve and hypoglossal nerve to innervate the **dura mater of the posterior cranial fossa.**
- Dorsal cervical plexus from **C1-3**
 - **Suboccipital n. C1** - Motor innervation to suboccipital muscles: rectus capitis posterior major, rectus capitis posterior minor, obliquus capitis superior and obliquus capitis inferior. C1 has no cutaneous nerve branches.
 - **Greater occipital n. C2** - *passes directly through the semispinalis capitis muscle* and provides sensory to the **posterior head.**
 - **3rd occipital n. C3** - Provides sensory innervation to the **posterior neck and lower part of the head**
- Nerves that innervate **dura**
 - Trigeminal → **Ophthalmic N., V1** → Dural branches: **Anterior and supratentorial dura**
 - Trigeminal → **Maxillary N., V2** → Dural branches: **Anterior and middle cranial fossa**
 - Trigeminal → **Mandibular N., V3** → Dural branches: **Middle and inferior cranial fossa; also sensory innervation to the TMJ**
 - **Vagus. X** → Dural/meningeal branches: **Dura of Posterior Cranial fossa**
- Structures in jugular foramen: internal jugular, CN IX, X, XI

- CN X related to N/V at jugular foramen
- Jugular foramen is between occipital and temporal bone

2. Discuss the muscles, bones, joints, fascial systems of the upper cervical spine and head.

- **Myodural bridge (complex)**
 - **OA membrane:** Superior attachment @ foramen magnum; inferior attachment @ posterior arch of C1
 - Dura mater firmly attach @ **C1-C3 and S2**
- **Trap & SCM = CN XI**
- **Scalenes** = ventral rami of **C4-6**
- **TMJ:** Innervation of the TMJ is from the mandibular branch of the trigeminal nerve.

3. Describe the role of the trigeminocervical ganglion in headache.

- **Trigeminocervical Nucleus - Integration**

The trigeminal nerve in the brain stem extends and becomes continuous with the grey matter of the spinal cord, becoming the trigeminocervical nucleus. The nucleus **receives afferent fibers from not only the trigeminal nerve, but from fibers of the VII, IX, and X cranial nerves and C1-3 spinal nerves**. This ramification **extends to the 3rd or fourth cervical segment** of the spinal cord. The significance of this nucleus is that all nociceptive afferent fibers to the head and neck exist in the specific area and have overlapping patterns of innervation with each other. This creates the basis for **referred pain in the head and neck**.

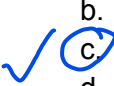
For example, pain in the head (CN V) can be referred back to a cervical or other cranial nerve segment resulting in pain in a different receptive field, or both fields

Radiology

1. Read a normal head CT and identify major cranial structures including foramina and their contents.
2. Determine proper imaging studies warranted for a patient with a headache
 - Imaging should be reserved for those patients who have red flags or who are not improving with standard medical therapy.
 - CT over MRI because of speed, cost and availability.
 - Non-contrast CT for acute stroke, severe acute headache and coma.
 - Looking for bleeding
 - Done first before other types of imaging to R/O bleed
 - Also good for identifying calcifications
 - CT with contrast if looking for a mass
 - MRI for visualizing specific soft tissue structures and vasculature (intracerebral pathology).
 - Most sensitive for finding a mass

Headache COAR Quiz

1. Which cranial nerve are you testing when having your patient puff out their cheeks with air while you try to push on their cheeks?

- a. CN V: V1 ophthalmic branch
- b. CN V: V2 Maxillary branch
-  c. CN VII
- d. CN III
- e. CN VIII

2. While observing for visual acuity and visual field testing, which cranial nerve are you assessing?

- a. CN I
- b. CN IV
- c. CN III
- d. Both CN II and III
- e. CN II

3. Which of the following CN's have associated parasympathetic fibers?

- a. III, V, X
- b. II, VII, IX, XI
- c. III, VII, IX, XII
- d. II, III, V, IX, X
- e. III, VII, IX, X

4. When performing the OM suture spread, which two cranial bones are you looking to release?

- a. Temporal, occipital
- b. Temporal, Parietal
- c. Occipital, Sphenoid
- d. Temporal, Sphenoid
- e. Occipital, parietal

5. When checking for tongue symmetry and even tongue motion, which cranial nerve are you assessing?

- a. CN X
- b. CN XII
- c. Both CN XI and CN XII
- d. CN IX
- e. CN XI

6. While observing for pupillary reaction (direct and consensual) you are assessing which cranial nerves?

- a. CN II & III
- b. CN III only
- c. CN II only
- d. CN II, III & IV
- e. CN III & IV

7. Which of the following are red flags for a patient with a headache?

- a. chills, phonophobia, tobacco abuse
- b. nausea, unilateral pain, cough
- c. Aura, obesity, fatigue
- d. fever, age>50, papilledema
- e. Photophobia, nausea, has had headaches like this before

8. You aim to check for an upper motor neuron disorder in your patient. You ask your patient to hold their arms outstretched with palms facing upward. What is the appropriate name for this test?

- a. Pronator drift
- b. Hoffman's reflex
- c. Corneal reflex
- d. Gag reflex
- e. Plantar response

9. What is the appearance of involuntary lifting of the legs when lifting a patient's head off the examining table with the patient lying supine called?

- ☒ a. Rovsing's sign
- ☒ b. Babinski sign
- ☒ c. Brudzinski sign
- d. Romberg sign
- e. Kernig sign

10. What is the best test to perform to rule out an intracranial bleed?

- ☒ a. non-contrast head CT
- b. lumbar puncture
- ☒ c. contrast head CT
- d. head MRI
- e. CBC

11. Your patient has a PMHX of hypertension and frequent migraines. Considering their comorbidities, which of the following drugs would be the best choice as your patient's prophylactic migraine medication?

- ☒ a. Valproate
- b. Amitriptyline
- c. Venlafaxine
- ☒ d. Timolol
- e. Topiramate

12. Which medication can be used as a migraine abortive?

- a. Propranolol
- b. Verapamil
- ☒ c. Sumatriptan
- d. Topiramate
- e. Botox

13. Nausea and vomiting with migraine may be related to tension on the dural sleeve involving which cranial nerve?

- a. CN VI
- b. CN IX
- ☒ c. CN X
- ☒ d. CN V
- e. CN XI

14. The trigeminocervical nucleus receives afferent nerve fibers from which cranial nerves, which may refer nociceptive pain to different areas in the head and neck?

- a. IX and X only
- ☒ b. V and X only
- c. X only
- d. V only
- ☒ e. VII, IX and X

15. Innervation of the scalene muscles is primarily from which of the following nerves?

- a. Transverse cervical nerve
- b. CN XI
- c. Dorsal scapular nerve
- ☒ d. Ventral rami of C4-C6
- ☒ e. Dorsal rami C1-C3

16. Which of the following muscles could the greater occipital nerve become compressed or restricted by?

- a. Splenius capitus
- b. Semispinalis cervicis
- c. Sternocleidomastoid
- ☒ d. Semispinalis capitus
- e. Longus capitus

17. The ophthalmic division of the trigeminal nerve passes through which cranial opening to reach the orbit?

- ☒ a. Supraorbital Foramen
- b. Stylomastoid foramen
- c. Infraorbital foramen
- ☒ d. Supraorbital fissure
- e. Foramen Ovale

18. The dural branches of maxillary nerve (V2) will innervate which of the following areas?

- ☒ a. Dura of the anterior and middle cranial fossa
- b. Superficial layer of spinal dura
- c. Dura of the middle and posterior cranial fossa
- d. Dura of the posterior cranial fossa
- e. Dura of anterior cranial fossae only

19. Sympathetic innervation to the face originates in which of the following spinal levels

- a. Occipitoatlantal region
- ☒ b. T1-T4
- c. C5-T1
- d. Trigeminal nerve
- e. C3-C5

20. The trigemino-cervical ganglion in the brainstem includes cell bodies from which of the following nerves?

- a. CN V, CN VII, CN VIII, CN X, CN XII
- b. C1-3, CN V, CN VII, CN VIII, CN XII
- ☒ c. C1-3, CN V, CN VII, CN IX, CN X
- d. C3-5, CN V, CN VII, CN IX, CN X
- ☒ e. CN III, CN V, CN X, CN XI, CN XII

21. Which of the following headaches would be characterized by unilateral, pulsing, "ice-pick" pain usually located around the eye or on one side of the head with possible associated symptoms of excessive eye tearing, or congestion, redness or sweating on the side of the pain?

- a. Cerebral Aneurysm
- ☒ b. Cluster headaches
- c. Occipital Neuralgia
- d. Tension Headache
- e. Migraine Headache- Common

22. Which of the following headache types would be characterized by acute onset of slurred speech, paresthesias in the arms or legs, dizziness and facial muscle weakness?

- a. Migraine Headache Common
- ☒ b. Stroke
- c. Migraine Headache Classic
- d. Cervicogenic Headache
- e. Tension Headache

23. Which of the following headaches may be due to the overuse of NSAIDs and is often described as a dull throbbing headache occurring daily, usually present in the morning?

- a. Occipital Neuralgia
- b. Nummular Headache
- c. Migraine Headache Common
- d. Rebound Headache
- e. Tension Headache

24. Occipital Neuralgia is characterized by:

- a. small, circumscribed area of continuous pain on an area of the head
- b. dull, pressure pain quality that wraps around the head near the temples
- c. Unilateral, pulsating, severe intensity pain associated with nausea or vomiting
- d. Unilateral sharp, stabbing pain with excessive lacrimation on the side of pain
- e. Shooting, Stabbing and sharp pain quality that originates at the base of the head and goes to the scalp

25. The patient in the case was found to have an SBS strain of left sidebending and rotation. Which of the following are true about this strain pattern?

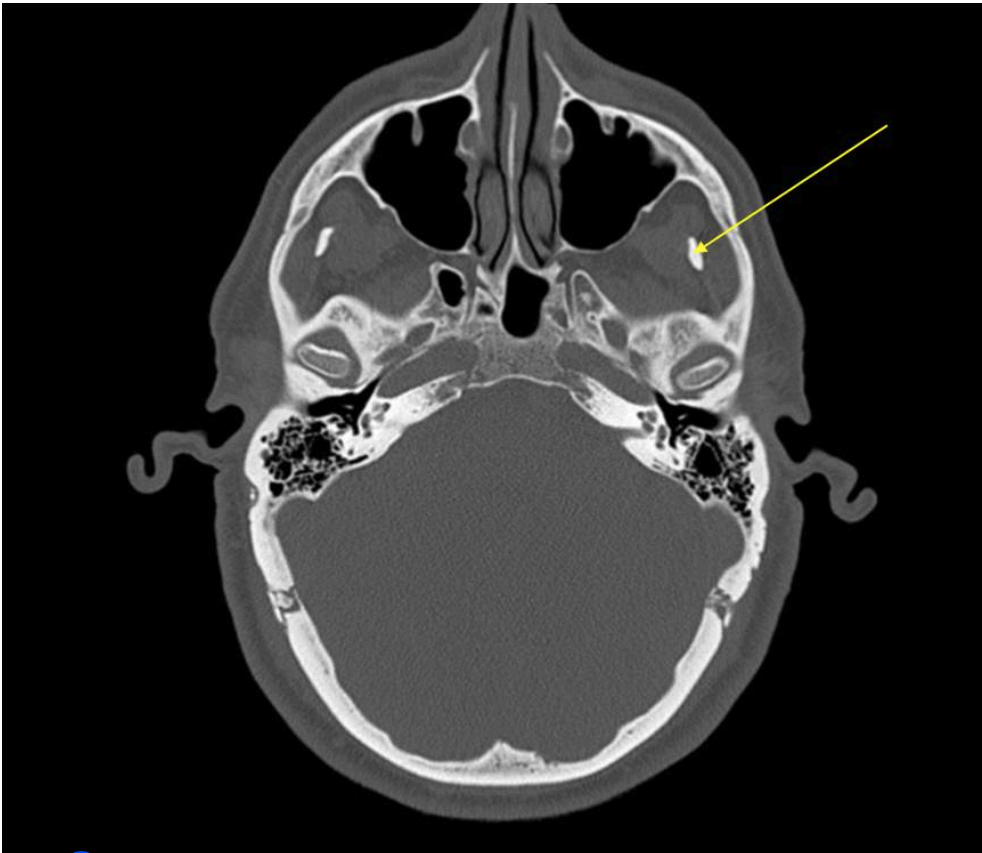
- a. There are two parallel vertical axes
- b. No motion in the head would be palpated
- c. There are two transverse axes
- d. There is one A/P and two parallel vertical axes
- e. There is one A/P axis

26.

What are your top 5 differential diagnoses for the patient presented to you in the COAR Headache Case?

Your Answer: *117 Migraine*

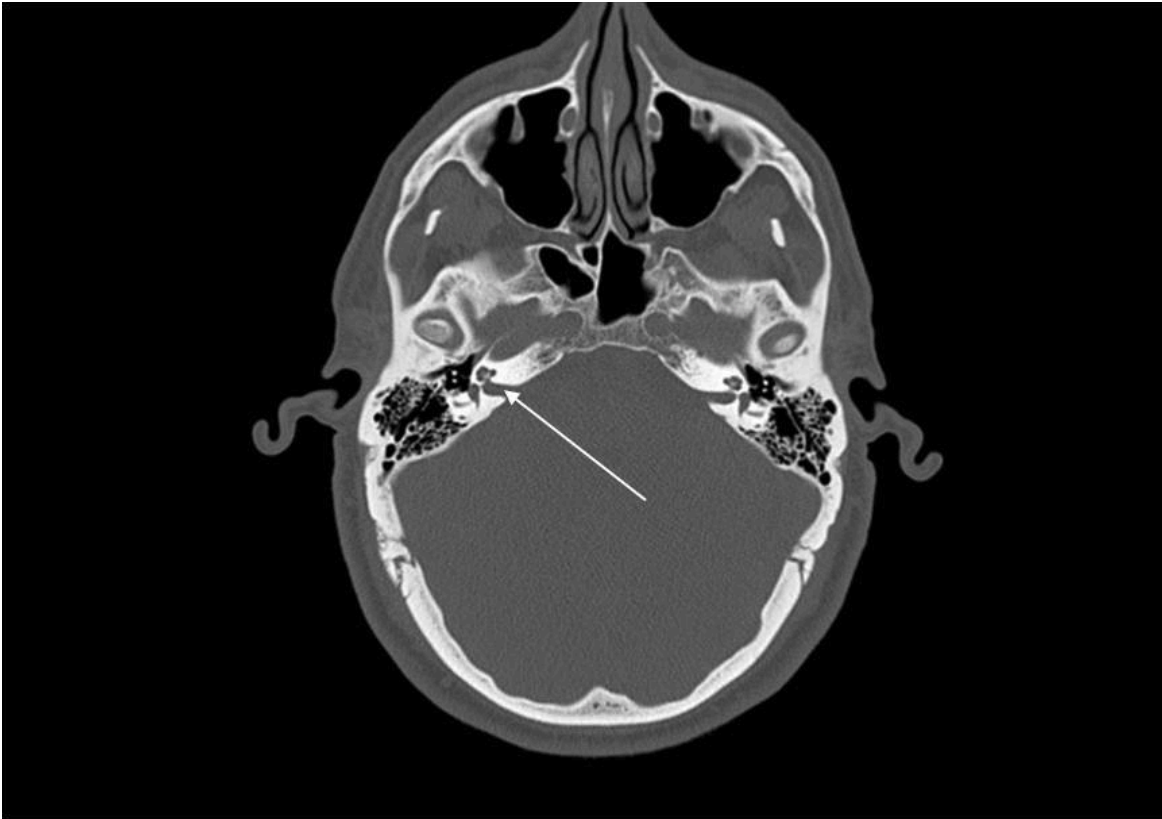
27. Which muscle has its attachment on the structure indicated by the arrow?



- a. Lateral pterygoid

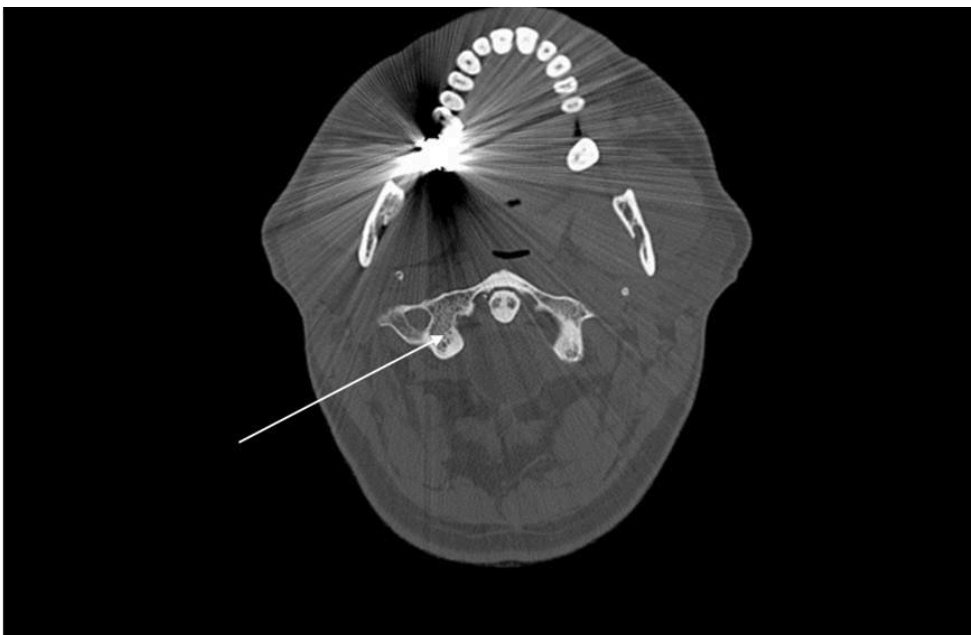
- b. Masseter
- c. Buccinator
- ☒ d. Temporalis
- e. Tensor veli palatine

28. Which of the following cranial nerve(s) exit the skull through the space indicated by the arrow?



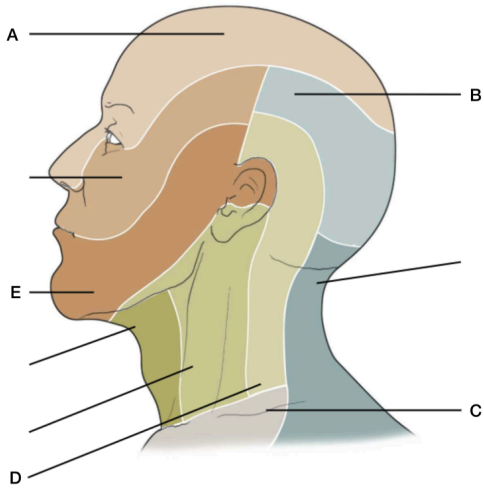
- ☒ a. VII & VIII
- ☐ b. IX, X, XI
- c. III, IV, VI, V1
- d. X
- e. V

29. Which nerve will exit superiorly to the vertebrae indicated by the arrow?



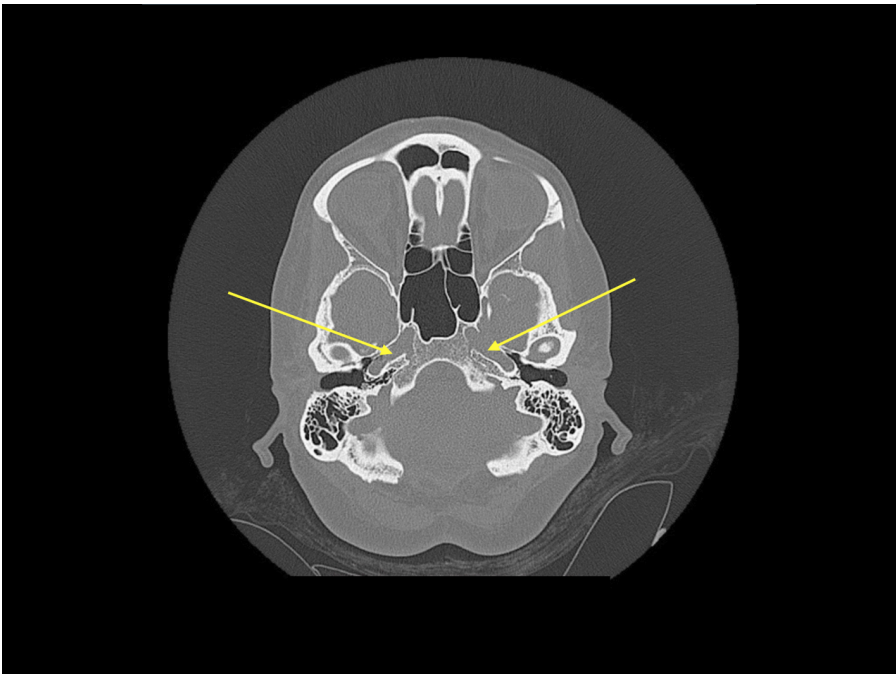
- a. Transverse cervical nerve
- b. Greater occipital nerve
- c. Suboccipital nerve
- d. Lesser occipital nerve
- e. Auriculotemporal nerve

30. The cutaneous innervation of the lesser occipital nerve is best represented by which of the following regions represented in the image?



- a. B
- b. C
- c. D
- d. A
- e. E

31. What structure occupies the space indicated by the yellow arrows?



- a. Middle meningeal artery
- b. CN VII and CN VIII
- c. CN IX, CN X, CN XI

- d. Internal carotid artery
- e. Trigeminal ganglion

Key:

- 1. C
- 2. E
- 3. E
- 4. A
- 5. B
- 6. A
- 7. D
- 8. A
- 9. C
- 10. A
- 11. D
- 12. C
- 13. C
- 14. E
- 15. D
- 16. D

- 17. D
- 18. A
- 19. B
- 20. C
- 21. B
- 22. B
- 23. D
- 24. E
- 25. D

26. Your own answer, one example:

- a. 1. Classic Migraine 2. Cervicogenic headache 3. Occipital Neuralgia 4. Post concussive syndrome 5. Transient Ischemic Attack

- 27. D
- 28. A
- 29. C
- 30. C (answer choice C, the answer is area D)
- 31. D

Week 9: Direct Study: Crossed Syndromes - Moloney

1. Describe the response of body tissues to gravitational strain pathophysiology (GPS)

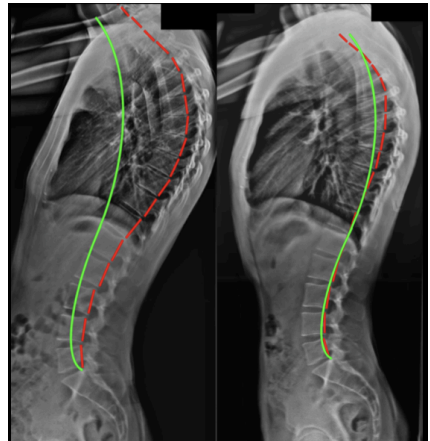
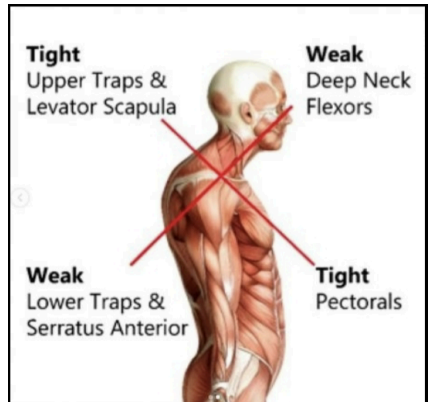
- **GPS** = Body tissues resist gravity effects leading to altered postural alignment & somatic dysfunction
 - Intrinsic Factors
 - Age: *muscle weakness, fascial elasticity, bone mass*
 - Altered integrity of soft tissue: *hypermobility, CT disorder*
 - Incompetent bony structure: *spondylolysis*
 - Poor Tissue Health: *disease, poor nutrition, smoking*
 - Altered base of support: *flat feet, high heels, short leg*
 - Change in body habitus: *pregnancy, weight gain*
 - Ergonomics
 - Extrinsic Factors
 - Surgeries
 - Trauma
 - Increased Gravitational Stress
- Tissue Response to GPS & Postural Decompensation
 - **Skeletal-Arthrodial Response** → Wolffe's Law → Bone remodels in response to stress
 - **Ligamentous Response** → Stabilizing Ligaments become stressed, tender, & calcified
 - Iliolumbar
 - Sacrotuberous
 - Long dorsal sacroiliac
 - **Muscular Response**
 - **Elastic:** Transient *functional* change in connective tissue length
 - **Plastic:** More permanent *structural* deformation due to postural changes
 - Fibrosis
 - Fatty Replacement
 - Myofascial remodeling along vectors of postural stress
 - Shortening non responsive to muscle spindle resetting (counterstrain)
 - Examples
 - **Trigger Points & Tendonitis** in weak/inhibited muscles
 - **Tears/Ruptures** in tight muscles
 - **Avulsion Fractures & Apophysitis** at attachment sites in tight muscles
 - **Sacroiliitis and Osteitis Pubis** from wear & tear
 - **Neural, Vascular, Lymphatic, and Visceral Element Response**
 - Decreased lymphatic/venous return at transition zones

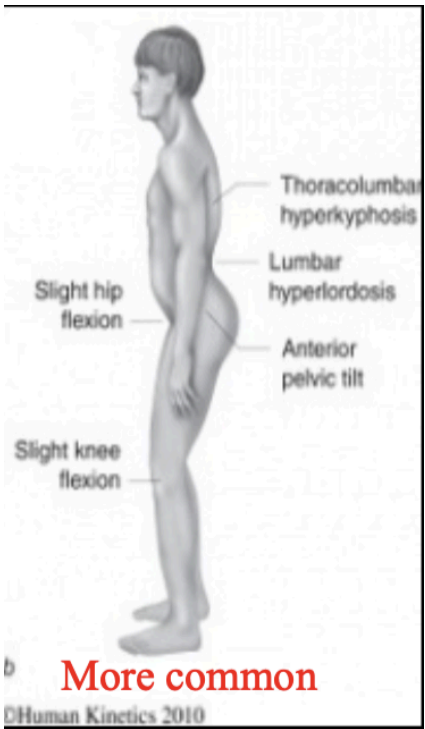
2. Identify the difference between a structural lesion and a functional lesion

Structural Lesions	Functional Lesions
- Focus on anatomy & biomechanics - Rely on imaging & damage to physical structures Diagnostic tests may be inconclusive due to functional cause	- Focus on cause of problem rather than pathology - Impairment of structure's ability to perform task that manifests as reflexive changes - Harder to diagnose & treat due to complex interactions within system

- Vertical adaptation** to a biomechanical lesion = effects on muscle firing
- Horizontal adaptation** = one joint problem leads to adjacent joints

3. Describe the components of the Upper and Lower Crossed Syndrome

Diagnostic Findings		
Posture/Viscera	Diagnostic Findings - Respiration - Thoracic Inlet - Thoracic Spine Proprioception Key Areas - Foot - SI Joint - C-Spine	
Upper Crossed	Diagnostic Findings - Forward Head Posture Extension of OA/C-Spine - Increased C/T Kyphosis Protracted, elevated, internally rotated shoulders	

Lower Crossed	Diagnostic Findings • Spinal Curves • Leg Lengths • Pelvic Tilt • Gait • LE Muscle Hypertrophy/Atrophy	 More common DHHuman Kinetics 2010 ^^Type A (tight hip flexors)^^
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4. Identify muscles involved in the upper & lower crossed syndrome (LCS) as postural/tonic or phasic

- Dynamic/Phasic: Smaller Muscle Groups
 - Fast Twitch
 - glycolytic
 - Respond by **inhibition, hypotonicity, weakness/pseudoparesis**
 - **Weak, hypotonic, weak with repetitive stretch**
 - Phylogenetically younger
 - Extensor Muscles
 - 2 Joint Muscles
 - Neck/Shoulder: **Deep Cervical Flexors, Mid/Low Trapezius, Serratus Anterior**
 - Hip/Pelvis: **Rectus Abdominis, Obliques, Transversus Abdominis**
 - LE: **Tibialis Anterior, Peroneal Muscles**
- Postural/Tonic: Larger Muscle Groups
 - Slow Twitch
 - oxidative
 - Respond by **facilitation, hypertonicity, shortening**
 - **Short, tight, facilitated w/ repeated stress**
 - Phylogenetically older
 - Flexor/Postural Muscles
 - 1 Joint Muscles
 - Neck/Shoulder: **Levator Scapulae, Upper Trap, SCM, Pectorals, Suboccipitals**
 - Hip/Pelvis: **Iliopsoas, Rectus Femoris**
 - Lumbar: **Erector Spinae**
- Most Common Agonist/Antagonist Pairings:
 - Biceps – Triceps
 - Deltoids – Latissimus Dorsi
 - Pectoralis Major – Trapezius/Rhomboids

- Iliopsoas – Gluteus Maximus
- Quadriceps – Hamstrings
- Hip Adductor – Gluteus Medius
- Tibialis Anterior – Gastrocnemius
- Anterior Deltoid – Levator Scapula
- Forearm Flexors – Forearm Extensors

5. Delineate the key components to treating muscle imbalance as outlined by Dr. Janda

- Sensorimotor system - sensory changes lead to altered motor program, neurologic and biochemical changes
 - Pain is the strongest stimuli
 - Altered proprioception >> altered afferent information >> altered CNS function >> effects motor output >> effects local joint >> effects global MSK response
 - Poor proprioception = recruit muscles harder
 - **Sensorimotor balance training** - first stage of exercise program
 - Dependent on vestibular, visual, and sensory systems
 - Look at excessive activity of knee, tibialis anterior, or toes → poor proprio
 - **Stretch before strengthening** - alleviate myofascial hypertonicity and relieve pseudoparesis
 - **Re-educate movement pattern** - correct firing pattern and strengthen weakened muscles
- Key Components
 - **Muscle Length:** Facilitation, tightness, shortening
 - **Muscle Strength:** Inhibition, pseudoparesis (inhibited antagonist due to tight agonist), weakness of phasic muscles
 - **Motor Control:** Loss of integrated coordinated neurologic function
- Key areas of proprioception for posture
 - **Foot**
 - **SI Joint**
 - **C-Spine**

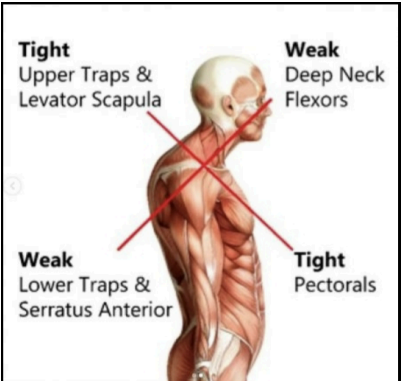
6. Outline the importance of addressing **faulty movement patterns** in the **lower** and **upper quarter**

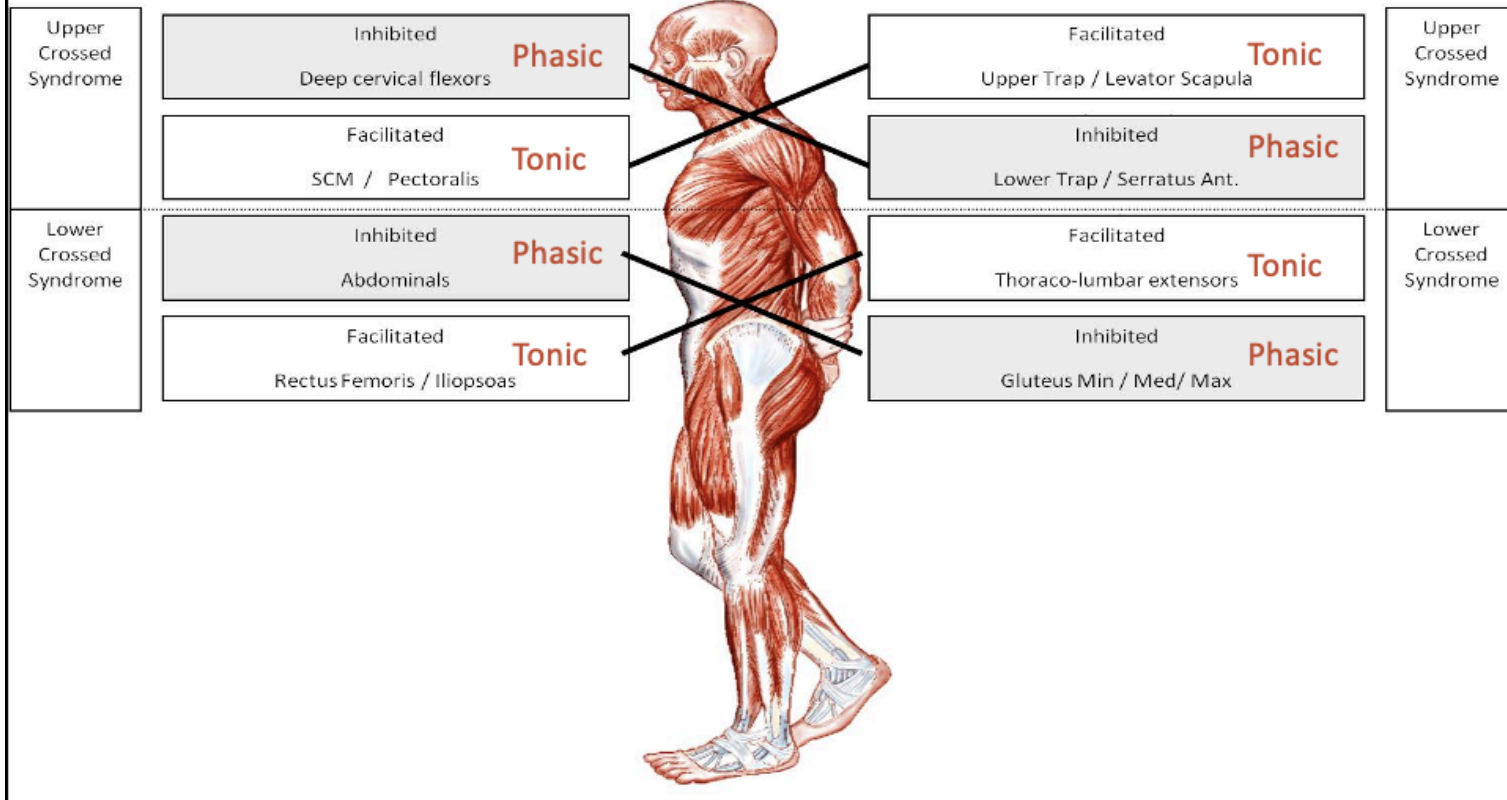
- Occurs in 2 major body regions: **pelvic** and **shoulder girdles**
- Altered principle muscle movements: **hip extension, hip abduction, shoulder abduction**
- UCS can result in
 - Brachial plexus restriction d/t to tight A/M scalenes, pec minor [tonic/facilitated]
 - Subacromial impingement d/t muscle imbalance of the rotator cuff mm. and deltoid, decreasing the SAS (i.e., functional impingement) → can lead to supraspinatus degenerative tears
 - Limited scapular rotation decreases SAS resulting in shoulder impingement d/t tonic pec minor and upper trap; phasic lower trap and serratus anterior

7. Review muscle imbalance and it's proposed causes

- Pathogenesis
 1. **Disturbance of muscular function** (attempt to return to homeostasis after posture alteration)
 2. **Afferent neural activity** → **cord, brainstem, cortex**
 3. **Efferent signal to alpha motor neuron** → **muscle contraction**
 4. **Muscle spindle adaptation** (γ system)
 5. **Muscle tone aberration** (d/t perpetuation)
- Neuro-reflexive component
 1. Altered proprioception
 2. Altered joint position
 3. Joint position becomes embedded in CNS as a motor pattern
 4. Chronic changes
- Causes
 - Postural adaptation to gravity
 - Neuro-reflexive due to joint blockage
 - Tight muscles need to be relaxed
 - Inhibited muscles need to be neurologically stimulated
 - Viscoelastic component
 - Tight muscles must be manually lengthened
 - CNS mal-regulation
 - Response to painful stimuli
 - Response to physical demands
 - Habitual movement patterns
 - Physiological influences
 - Lifestyle: stress, fatigue, insufficient movement, repetitive movements, bracing, anticipation of pain

8. Apply understanding of the LCS and UCS to diagnostic findings and proper treatment given those findings

Diagnostic Findings		
Upper Crossed	Diagnosis • OA, AA, C2-7, T-Spine, Ribs • Protracted, elevated, internally rotated shoulders Diagnostic Findings • Upper neck, back, shoulder pain • Shoulder Impingement Syndrome • Thoracic Outlet Syndrome • Carpal Tunnel Syndrome • TMJ Dysfunction • Chronic Headaches • Recurrent 'knots' or trigger points Treatment • Sensorimotor/Proprioception Balance Testing • Stretch Upper Body Facilitated/Tonic Muscles using Muscle Energy ○ Levator Scapula ○ Upper Trap ○ SCM ○ Pec Major/Minor ○ Latissimus dorsi ○ Posterior shoulder capsule • Re-education of movement pattern of weak phasic muscles ○ Cervical Flexion retraining ○ Rhomboid & Lower Trap	



Lower Crossed

Diagnosis

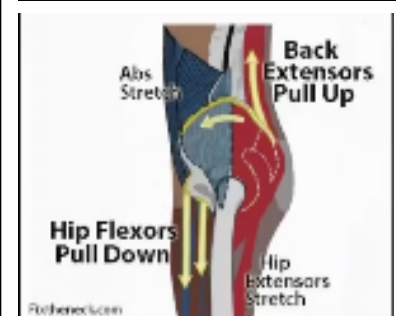
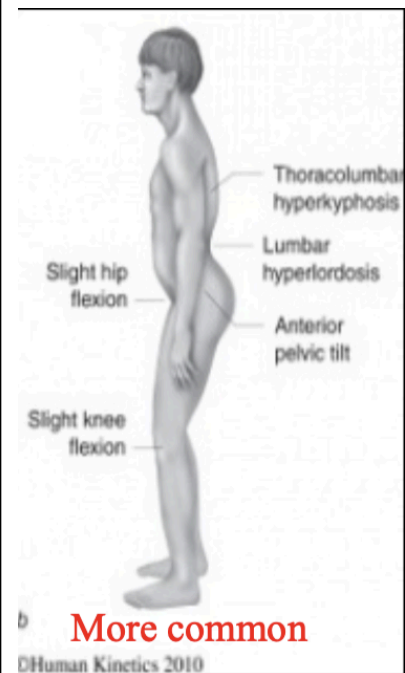
- Mitchell Model
- **Type A LCS [more common]** = due to **tight hip flexors**
- **Type B LCS** = due to **weak abdominal wall**

Diagnostic Findings

- Spinal Curves
- Leg Lengths
- Pelvic Tilt - **typically anterior pelvic tilt**
- Gait
- LE Muscle Hypertrophy/Atrophy

Treatment

- Sensorimotor/Proprioception Balance Testing
- Stretch Lower Body Facilitated/Tonic Muscles with **Muscle Energy**
 - Iliopsoas
 - Rectus Femoris
 - Tensor Fascia Lata
 - Hip Adductors
 - Piriformis
 - Hamstrings
 - Quadratus Lumborum
 - Gastrocnemius/Soleus
- Re-education of movement pattern/strengthening of weak inhibited/phasic muscles



9. Identify the 6 most common dysfunctions in “failed lower back pain syndrome”

- Non-neutral (type 2) lumbar dysfunction
- Pubic symphysis dysfunction
- Sacral base posterior
- Innominate shears
- Leg length discrepancy with un-level sacral base
- **Muscle imbalance [most prevalent - 90%]**

Know for Lab and for the TEST!

• Function of (should know origin/insertion too but will not be tested on it):

- Upper Trapezius- elevates the scapula
- Lower/Middle Trapezius-retracts and rotates the glenoid superiorly and stabilize the scapula
- Levator Scapulae- elevates the scapula
- Scalenes- head sidebending and some rotation
- Sternocleidomastoid- flexes the c-spine and sidebending and rotation
- Pectoralis Major-shoulder adduction
- Pectoralis Minor- shoulder adduction and some flexion
- Deltoid- abduction, flexion and extension of the shoulder
- Rhomboids- scapular retraction
- Serratus Anterior- scapular protraction
- Subscapularis- shoulder internal rotation
- Infraspinatus- shoulder external rotation
- Supraspinatus- shoulder abduction to 15 degrees
- Deep Neck Flexors (Longus Colli, Longus Capitus)- cervical flexion
- Latissimus Dorsi-extends, adducts humeral, rotates scapula inferiomedially.

Know for Lab and for the TEST!

Function of (should know origin/insertion too but will not be tested on it):

- Iliopsoas- hip flexion
- Rectus Femoris- hip flexion
- Tensor Fascia Lata- hip flexion, IR, ABD
- Lumbar Erector Spinae- spine extension, minor hip extension
- Quadratus Lumborum- ipsilateral sidebending, spine extension, minor hip extension
- Abdominals- Spine flexion
- Glut Max- hip extension
- Glut Medius and Minimus- hip ABD/IR
- Hip Adductors- hip Adduction
- Piriformis- Hip ER, above 90 degrees abduction
- Quadriceps (3 vastus)- knee extension
- Rectus femoris- Knee extension, hip flexion
- Gastroc/Soleus- Ankle plantar flexors
- Tibialis Anterior- Ankle dorsiflexor

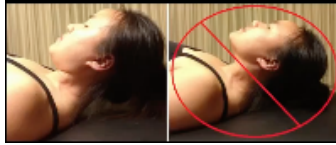
UCS & LCS Treatment

- The Janda Recipe
 - Sensorimotor balance training
 - Stretch before strengthen
 - Re-educate movement pattern
- Treatment Goal
 - Restore normal posture & movement
 - Alleviate pain
 - Protect osteoarticular system
 - Reduce strain on joints & ligaments

UCS Treatment

Cervical Flexion Test

- Pt Supine
- Have pt slowly look to feet
- Forehead should be more anterior than chin through ROM

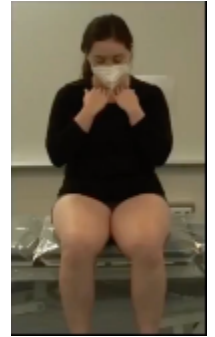


Significant Findings

- Weak deep cervical flexors (Phasic)
- Hypertonic SCM, Scalenes, Cervical Erector Spinae

Cervical Flexion Retraining

- Pt palpates above clavicle onto anterior cervical muscles
- Pt nods head to engage deep neck flexors without contracting superficial anterior cervical muscles
- Hold 5-7 secs



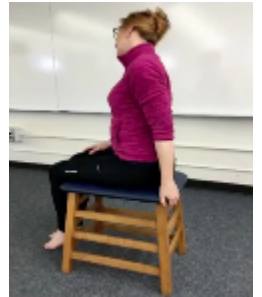
Bilateral Shoulder Abduction Test

1. Have pt abduct arms
2. Observe which scapula is more protracted and elevates earlier

Significant Findings if pt sidebends t-spine & elevates shoulder to achieve abduction

LS/SCM/Upper Trap Self Stretch

- Pt sitting, hold onto edge of chair
- Sidebend away and rotate/tuck chin away until stretch felt
- Modify to target different fibers



Scapular Stabilization

1. Pt on hands n knees
2. Observe for winging or elevation
3. Have pt lift one arm off while observing supporting arm



Findings could be due to weak rhomboids, serratus, lower traps or tight pecs, upper traps, levator scapula

Pec Self Stretch

- Pt hand on wall & slowly turn away from side u want to stretch

Muscle Retraining: Rhomboid & Lower Trap

- Pt interlaced hands in front and pull elbows backwards >> rhomboids
- Pt interlaced hands above head and pull elbows down >> lower traps

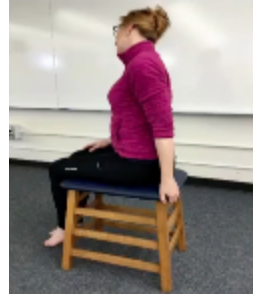
Scapular Depression

LS/SCM/Upper Trap Self Stretch

1. Pt prone and arm flexed
2. Monitor inferior medial border
3. Look for contraction of lower trap and have pt pull scapular down against resistance



- Pt sitting, hold onto edge of chair
- Sidebend away and rotate/tuck chin away until stretch felt
- Modify to target different fibers



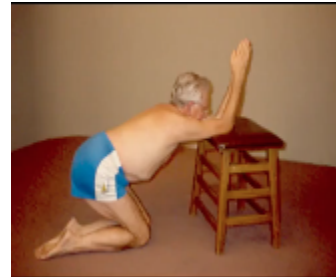
Supine Bilateral Shoulder Flexion

- Pt supine
- Bring pt hands overhead into shoulder flexion
- Assess tension & ROM & ability to maintain neutral spine



Lat Self Stretch

- Pt kneel and elbows on stool or table
- Pt drops chest to floor and sit back on heels



LCS Treatment

Sensorimotor/Proprioception Balance Testing

- S1: One Leg, Arm Down, Eye Open
- S2: One Leg, Arm Crossed, Eye Open
- S3: One Leg, Arm Down, Eye Closed
- S4: One Leg, Arm Crossed, Eye Closed

Pelvic Clock:

- Tilt Pelvis Posterior > 12 o clock
- Tilt Pelvis Anterior > 6 o clock
- Rotate Pelvis Right > 9 o clock
- Rotate Pelvis Left > 3 o clock

Pelvic Tilt w /Abduction test

- Pt bends knees and keeps feet together
- Pt tilt pelvis posterior in 12 o clock position
- Pt keeps feet together and slowly abducts knees
- Fail if ASIS moves while legs abducting, sign of weak abdominals

Pelvic Tilt w Heel Side

- Pt bends knees and keeps feet together
- Pt tilt pelvis posterior in 12 o clock position
- Pt keeps feet together and slowly extends leg with heel on table
- Fail if ASIS moves while legs extending, sign of weak abdominals

Hip Extension Firing Pattern

1. Hamstrings
2. Gluteus Maximus
3. Contralateral Lumbar Erector Spinae
4. Ipsilateral Lumbar Erector Spinae

Significant Findings: Delayed/absent firing of glutes and substitution by hamstrings and erector spinae

Gluteus Maximus Strengthening



Hip Abduction Firing Pattern

1. Gluteus Medius
2. Tensor Fascia Lata
3. Quadratus Lumborum
4. Lumbar Erector Spinae



Significant Findings: Late firing of gluteus medius and early firing of TFL >> internal rotation & flexion of the hip

Retraining of glut med/min



Quadratus Lumborum Assessment

1. Pt lies on contralateral side of QL being assessed
2. Stabilize 12th rib
3. Abduct Leg to relax QL, then allow leg to drop to assess QL length
4. Compare both sides
5. Utilize ME by having pt abduct leg while providing resistance to gain QL length between 12th rib and iliac crest

QL Self Stretch

- Im a little teacup short n stout pose

Hamstring Assessment

- Pt supine
- Lift pt leg up to assess hamstring hypertonicity
- Do both sides

Hamstring Self-Stretch

- Pt supine
- Pt lift leg and use towel around bottom of foot to induce stretch

Adductor Assessment

Adductor Self Stretch

Crossed Syndromes Quiz

1. Which of the following is NOT part of the muscle imbalance description?

- ☒ a. Muscle Size
- b. Muscle Length
- c. Muscle Strength
- ☐ d. Motor Control

2. Which of the following is true of Phasic Muscles?

- ☒ a. Respond by inhibition
- b. Respond by facilitation
- c. Large muscle groups
- ☐ d. Respond by hypertonicity
- e. Typically shorten

3. Which of the following is true of upper crossed syndrome posture?

- ☒ a. OA flexion
- ☒ b. Decreased thoracic kyphosis
- ☒ c. Protracted scapula
- ☒ d. Retracted head posture
- ☒ e. Externally rotated shoulders

4. In the lower crossed syndrome, which of the following muscles is typically tight and facilitated?

- a. Vastus Medialis
- ☒ b. Iliopsoas
- ☒ c. Gluteus Medius
- ☒ d. Abdominal muscles
- ☒ e. Gluteus Maximus

5. Which of the following muscles is typically weak and inhibited in the lower cross syndrome?

- ☒ a. Quadratus Lumborum
- ☒ b. Erector spinae group
- ☒ c. Adductor longus
- ☒ d. Tibialis anterior
- ☒ e. Gastrocnemius

6. Which of the following describes a PHASIC muscle?

- ☒ a. Phylogenetically younger
- b. Generally flexors
- ☒ c. Oxidative/slow twitch
- ☒ d. Typically one joint muscles
- e. Become short, tight and facilitated with stress

7. Which of the following typically becomes tight and facilitated in the upper crossed syndrome?

- ☒ a. Rhomboids

- ~~b. Middle and lower trapezius~~
- ~~c. Deep Cervical Flexors~~
- ☒ d. Levator Scapulae
- ~~e. Infraspinatus~~

8. Which of the following typically becomes weak, inhibited and hypotonic in the upper crossed syndrome?

- a. Scalenes
- b. Subscapularis
- ☒ c. Supraspinatus
- d. Flexors of the upper extremities
- ~~e. Pectoralis~~

9. In order to improve shoulder impingement symptoms, what shoulder imbalance would you want to address?

- a. Strengthen the rotator cuff
- ~~b. Strengthen pectoralis major~~
- ☒ c. Strengthen Lower Trapezius
- ~~d. Strengthen Upper Trapezius~~
- ~~e. Strengthen Levator Scapulae~~

10. Who is most associated with the concept of crossed syndromes?

- a. Andrew Taylor Still
- b. William Garner Sutherland
- ☒ c. Robert Fulford
- ☒ d. Vladmir Janda
- e. Viola M. Frymann
- f. <3 Gar Singer <3
- g. <3<3<3 Haaris Hameed <3<3<3
- h. Victor Nuno (the goat)

KEY

- 1. A
- 2. A
- 3. C
- 4. B
- 5. D
- 6. A
- 7. D
- 8. C
- 9. C
- 10. D (F depending on who you ask because he's a walking upper and lower cross specimen)

Cumulative material (per Dr. Wolf, OCMM relating to headache COAR)

OCMM 1 - Wolf

Objectives:

1. Identify the five (5) components of the Primary Respiratory Mechanism (PRM) and their interrelationship.

1. **Brain & Spinal Cord has inherent motility (CNS)**

2. **CSF has rhythmic fluctuations → moves dura → moves cranial bones**

- a. Immunologic Purposes
- b. Metabolic/Endocrine Purposes

3. **Intracranial and Dura have inherent mobility**

- a. Mobility dependent on CNS motility & CSF fluctuations
- b. Reciprocal Tension Membrane:
 - i. Dura attaches at **C2, C3, Foramen Magnum, and Sacrum (S2 onward)**
 - ii. Cranium & Sacrum are interconnected
 - iii. What happens in the cranium, also happens in the sacrum, and vice versa

4. **Osseous cranial bones have articular mobility**

- a. Sutures are patent into 80s
- b. Living flexible bone designed for motion

5. **Mobility of sacrum between ilia secondary to core link from dura**

2. Differentiate between motility and mobility and the different functions of each within the PRM.

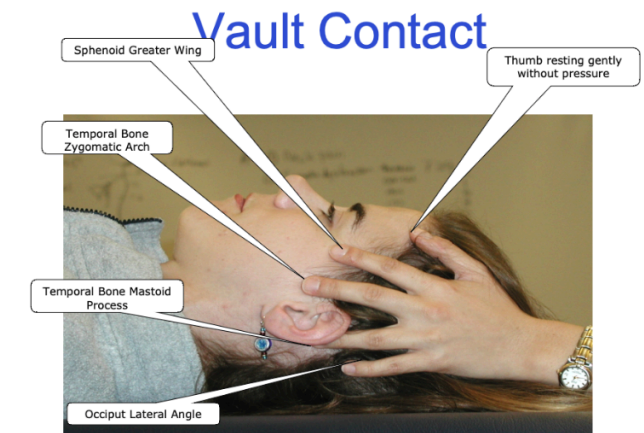
- Motility: Independent Movement
- Mobility: Dependent on other movements

3. Explain the concept of the core link, and understand its importance to somatic dysfunction in the cranium and beyond.

- Superiorly, dura attached at **Foramen Magnum and C2, C3** (myodural bridges)
- Inferiorly, dura is strongly attached to **S2** on sacrum (superior transverse axis) and **coccyx**
- Motion in head will trigger motion in sacrum & vice versa

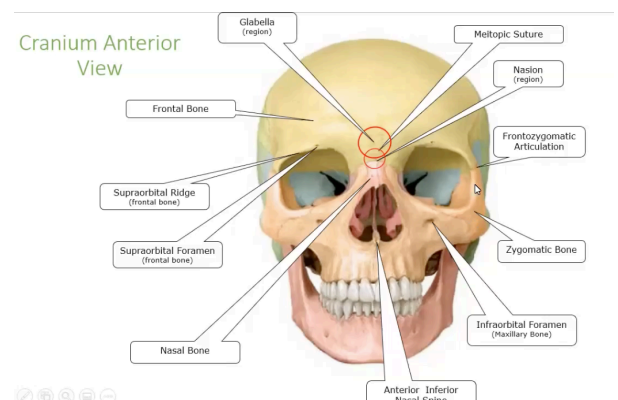
4. Identify the vault contact and the landmarks that each finger contacts in this position.

- Thumb: Wherever it lands, but gently
- Digit 2: Greater Wing of Sphenoid Bone
- Digit 3: Zygomatic Arch
- Digit 4: Mastoid Process
- Digit 5: Lateral Angle of Occiput



OCMM1 Lecture 2

1. Identify emphasized adult cranial occipital bone anatomy, landmarks, and articular Associations



a. Midline Bones (**SOVE**): Flex/Extend

- i. Sphenoid
- ii. Occiput
- iii. Vomer
- iv. Ethmoid (Median Plate)

b. Paired Bones: Internal/External Rotation

- i. Ethmoid (Lateral Masses)
- ii. Frontal
- iii. Maxilla
- iv. Parietals
- v. Temporals
- vi. Zygoma
- vii. Lacrimals
- viii. Nasal

c. **Glabella**: Between Orbital Ridges on Frontal Bone

d. **Nasion**: Region where frontal bone meets nasal bone

e. **Bregma** (Anterior Fontanelle): Where coronal & sagittal sutures overlap

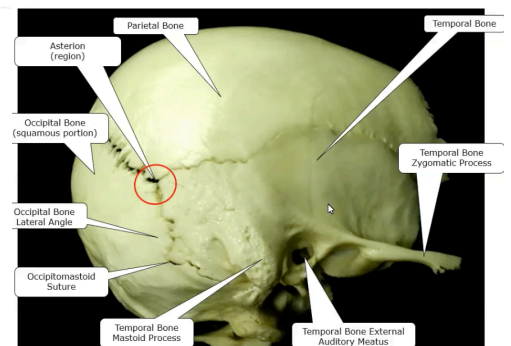
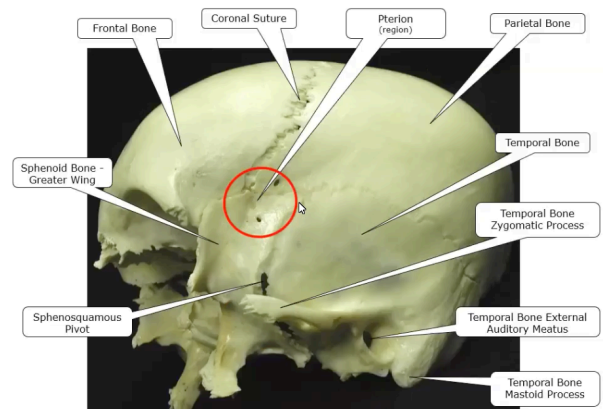
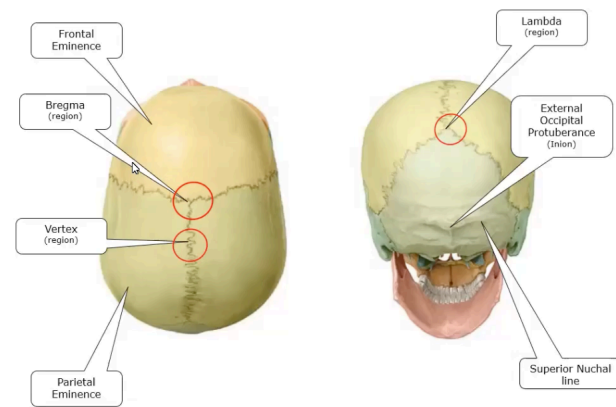
f. **Vertex**: where they hang up skeletons lmao where parietals meet

g. **Lambda** (Posterior Fontanelle): where sagittal + lambdoidal suture overlap

h. **Pterion**: Frontal, Sphenoid, Temporal, Parietal meeting point

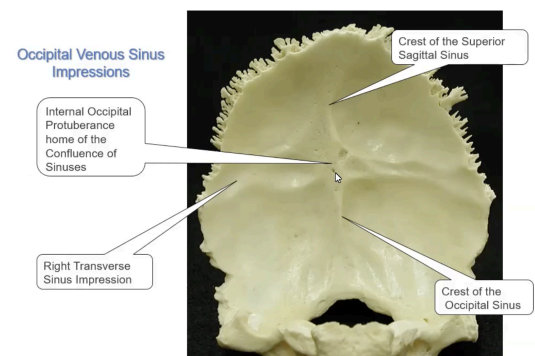
i. **Sphenosquamous Pivot Point**: Junction between squamous portion of temporal with sphenoid bone

j. **Asterion**: Parietal, Temporal, Occipital Meeting Point



2. Identify the following dural sinuses' developmental impression(s) and associated locations with the occipital bone:

- a. Superior Sagittal Sinus
- b. Transverse Sinuses
- c. Confluence of Sinuses
- d. Occipital Sinus
- e. Straight Sinus
 - i. imprint of because it's in dura not on bone

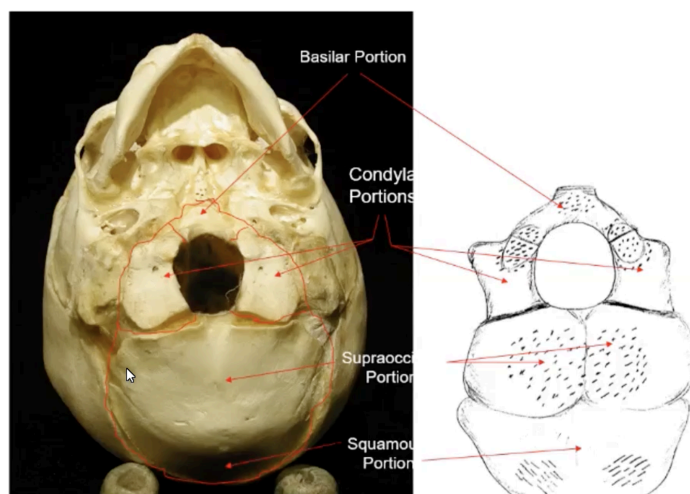


3. Identify the location of the sphenoid and occiput relative to their central position in the skull, and palpatory points when utilizing the Vault contact

•

4. List the four (4) developmental parts of the occipital bone that are present at birth - appreciating the role the cartilaginous connections play in birth trauma

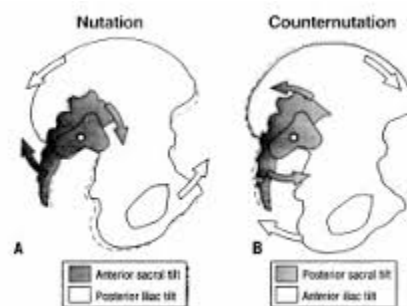
- **Squamous portion with subparts**
 - *Interparietal* — *membranous*
 - *Supraocciput*
 - **Two Condylar Portions**
 - Hypoglossal Groove
 - Anterior lateral aspect of foramen magnum
 - Forms posterior $\frac{2}{3}$ of condyles
 - **Basilar Portion**
 - Articulates with SBS
 - Forms anterior $\frac{1}{3}$ of condyles
- Note: Fuse at 7-8 years old



OCMM1 Lecture 3

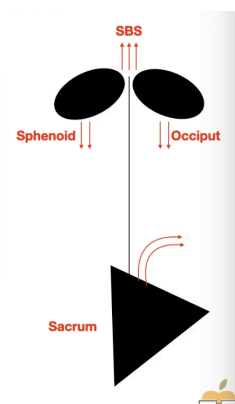
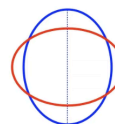
Describe how PRM influences inhalation/exhalation, flexion/extension, internal/external rotation, and how these are interrelated.

- **Inhalation**
 - CNS rhythmic **expansion** and uncoiling
 - Midline bones in **flexion**
 - Paired bones in **external rotation**
 - **SBS Rises**
 - **Counternutation of Sacrum** (Biomechanical Extension)
- **Exhalation**
 - CNS rhythmic **contraction** and coiling
 - Midline bones in **extension**
 - Paired bones in **internal rotation**
 - **SBS Falls**
 - **Nutation of Sacrum** (Flexion)
- **Flexion**
 - Elevation of SBS
 - Sphenoid rotates anteriorly
 - Occiput rotates posteriorly
 - Paired Bones External Rotation
 - Ernie
 - Shortening of A-P Diameter (Blue → Red)



Flexion

Sphenobasilar Synchondrosis	Cephalad
Sacrum	Posterior & Counternutates
Respiration	Inhalation
Paired Bones of the Cranium	Externally Rotate
AP Diameter of the Cranium	Shortens



- **Extension**
 - Depression of SBS
 - Sphenoid rotates posteriorly

- Occiput rotates anteriorly
- Paired Bones Internal Rotation
- “Bert”
- Lengthening of A-P Diameter

Dysfunction	Axis/Axes	Coupling	Naming Convention
<i>Torsion</i>	1 AP	Rotate OPPOSITE	Greater wing of the sphenoid
<i>Sidebending/Rotation</i>	2 Vertical 1 AP	Rotate SAME (AP) Rotate OPPOSITE (Vertical)	Convexity in sidebending
<i>Vertical Strain</i>	2 Transverse	Rotate SAME	Base of the sphenoid
<i>Lateral Strain</i>	2 Vertical	Rotate SAME Translate OPPOSITE	Sphenoid translation
<i>Compression</i>	-----	Sphenoid & Occiput move toward one another	-----

- Internal Rotation
 - SBS falls
 - Exhalation Phase
- External Rotation
 - SBS rises
 - Inhalation Phase

1. Describe the physiologic PRM phasic motion of the sphenoid and occiput in a “freely mobile” cranium. This includes:
 - a. Location and orientation of physiologic PRM axes of motion
 - b. Sphenobasilar Synchondrosis (SBS) PRM motion description
 - There are dural attachments at foramen magnum
 - As SBS rises (flexion), this leads to counternutation of sacrum (extension)
 - As SBS depresses (Extension, this leads to nutation of sacrum (flexion)
 - c. Vault contact palpatory “air head” perception of the sphenoid greater wings and occipital angles undergoing the physiologic phases of the PRM

OCMM 5 - Clinical Considerations 1 - Sloane

Learning Objectives:

1. Identify the emphasized cranial, muscular, ligamentous, & neural anatomy related to the pathophysiology of the following clinical cases:

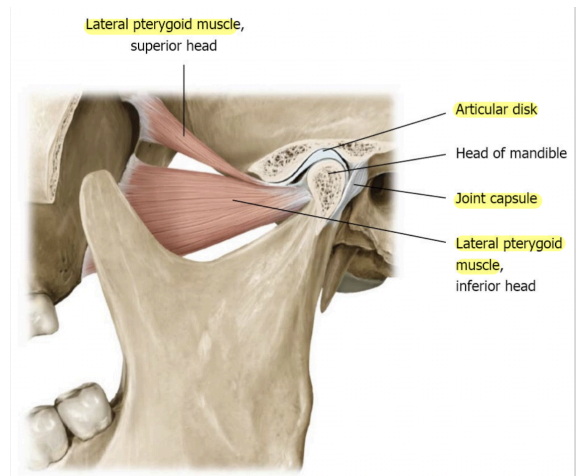
- **Headache**
 - **Trigemincervical nucleus** allows for crossover, creating referred pain — close association with cervical spinal nerves enables tx at the cervical spine to relieve headaches
 - **C1-C3** and the **Trigeminal Spinal Tract V, VII, IX, and X**
 - T1-T4 Sympathetic Innervation to Head and Neck
 - Suboccipital Muscles
 - Meckels' cave (dural pouch that houses **trigeminal ganglion**)- susceptible to dural strains
 - Temporal & Sphenoid Bone Somatic Dysfunction (petrosphenoidal articulation - Meckel's Cave)
 - Superior orbital fissure (sinus HA):
 - V1: Sensory **Anterior** Cranial Fossa
 - Sphenosquamous articulation:
 - V2 (F. Rotundum): Sensory **Anterior and Middle** Cranial Fossa
 - Dura including **falx cerebri and tentorium cerebelli** (mainly innervated by V1)
 - V3 (F. Ovale): Sensory **Middle** Cranial Fossa
 - CN X (Jugular Foramen) + C-Spine Fibers innervate **Posterior** Cranial Fossa
 - Pelvis and LE: reciprocal compensatory patterns in dura (RTM)
 - Spine (Scoliosis): Upper cervical pathology can lead to headaches
 - Assessing Normal Motion
 - Temporal Bones (Paired Bone) PRM:
 - Internal Rotation (Exhalation)
 - External Rotation (Inhalation)
 - Sphenoid/Occiput PRM
 - Flexion
 - Extension
- **TMJD**
 - Four categories
 - **Myofascial pain dysfunction (MPD) syndrome** — MOST COMMON
 - muscles of mastication
 - weakness in deep cervical flexor muscles
 - **Internal derangement (ID) injury**
 - abnormal relationship between **articular disc** and mandibular condyle
 - acute disc displacement and chronic recurrent dislocation
 - **Accessory ligament dysfunction (ALD)**
 - Accessory **ligaments** that help stabilize the joint
 - sphenomandibular, stylomandibular, and pterygomandibular
 - **Degenerative joint disease (DJD)**
 - Organic **degeneration of the articular surfaces** within the TMJ

- **TMJ Joint** = convex articular surface anteriorly and concave posteriorly; articular disc in biconcave (three parts)

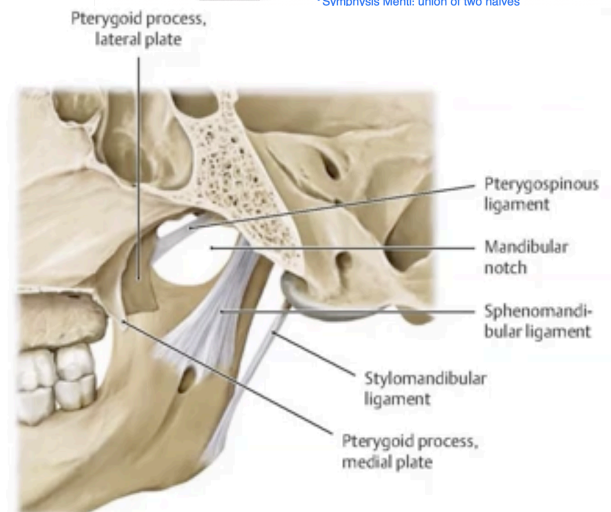
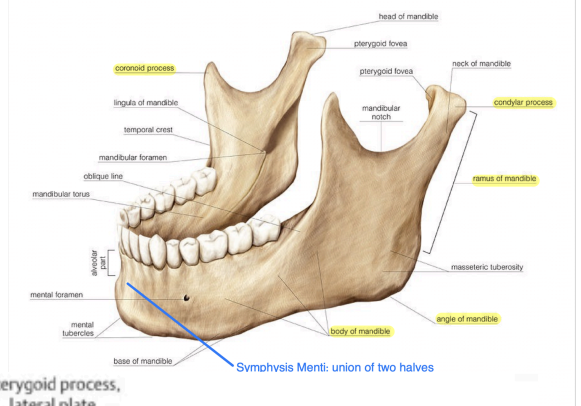
- Hinge joint w/ gliding motion
- Motion = **suprahyoid muscles** (opening), **lateral pterygoid** (gliding), **temporalis**, **masseter**, and **medial pterygoid** (closing)
- Anatomy:
 - compound synovial joint
 - contains non-vascular fibrous connective tissue
 - fibrocartilaginous surface on TMJ
 - articulation involves a fibro-cartilaginous surface

- **TMJ Disc** = articular disc divides joint into upper and lower compartments

- Disc is thinner in middle vs. ends
- Disc moves forward with condyle of mandible
- Lateral pterygoid attaches to disc
- Motion



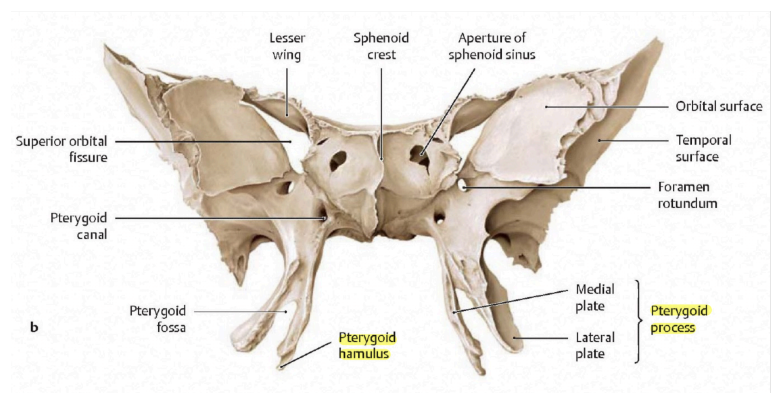
*Not a part of either neurocranium or viscerocranium- develops from membranous ossification.



- **TMJ Ligaments**

- Stylomandibular Ligament
- Sphenomandibular Ligament
 - Slack when TMJ is CLOSED
 - Taut when TMJ is OPEN
- Pterygospinous Ligament

- Somatic Dysfunctions found in distant regions with TMD

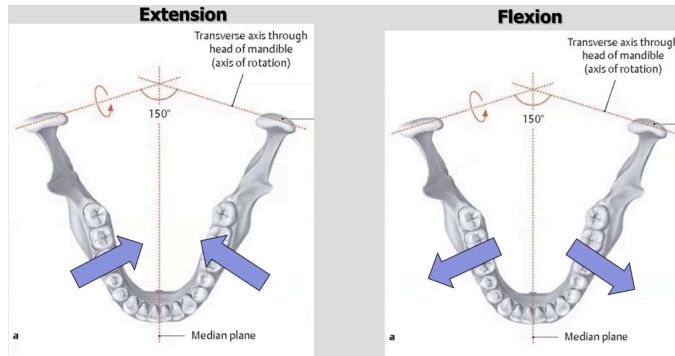


- Cranial dysfunction of some kind - nearly all
- OA - 9%
- C3-C4 dysfunction - 50%
- C4-C5 - 30%
- T2-T3 - 14%
- T3-T4 - 14%
- Lumbosacral junction - 29%
- Sacroiliac joints - 32%
- Scoliosis - 14%
- Myofascial Pain Referral to Muscles of Mastication
 - **Masseter** (CLOSES JAW): trigger point symptoms - upper/lower teeth pain, maxillary pain, **ear pain, tinnitus**, reduced jaw opening
 - **Lateral pterygoid** (OPENS JAW): unilateral contraction causes **deviation to one side**
 - inserts onto intra-articular disc
 - Signs/Sx = **reduced opening**, altered occlusion, maxillary region pain, TMJ pain, clicking
 - **Medial pterygoid** (CLOSES JAW): unilateral contraction causes **deviation to opposite side**
 - Signs/Sx = mouth and throat pain, painful swallowing, “stiffness” related to ear
 - **Temporalis** (Closes Jaw)
 - Signs/Sx = **Zig-Zag Deviation**, Headache, Upper Teeth Pain
- Popping/Clicking
 - Anterior displacement of disc relative to condyle
 - Jaw Open: condyle settles on central portion of disc, causes click
 - Jaw Close: condyle relocates behind disc, causes reciprocal click
- Jaw Locking: Condyle cannot locate onto disc or pushes disc forward when attempting to open jaw
- **Vertigo/Dizziness**
 - Benign Paroxysmal Positional Vertigo
 - **Debris in endolymph** of **semicircular canals**, last 30 sec-2 min and resolved with repositioning
 - **NO hearing loss or tinnitus**
 - Cervicogenic Vertigo = dizziness, unsteadiness, unbalance of whiplash injury
 - **concurrent with neck** pain, stiffness, worse with neck movement
 - **NO hearing loss or tinnitus**, possible nausea
 - disturbance in cervical proprioceptive input to vestibular nuclear complex and cerebellum
 - Meneire's Disease
 - Congestion in the membranous labyrinth, **blockage of the endolymphatic duct**
 - Severe vertigo; **sometimes permanent tinnitus and hearing loss**
- **Bell's Palsy**
 - Inflammation of facial nerve within **osseous facial canal (temporal bone)**, causing compression and ischemia of cranial nerve VII
 - Signs/Sx = unilateral hemiparesis of the upper/lower face, change in taste (anterior 2/3 tongue), hyperacusis (stapedius), decreased lacrimal & corneal reflexes (not emotional tears)
 - Anatomy = **To Zanzibar By Motor Car** (Temporal, Zygomatic, Buccal, Mandibular, Cervical)

2. Define the role of different treatments, including OCMM and other OMT techniques, as it relates to the pathology discussed and emphasized in the self-study

● TMJD

- **Acute:**
 - consider OMT *before and after* dental work
 - often *self-limiting*
 - patient education = self massage, exercises, mindful of chewing, etc.
- **Chronic:**
 - OMT
 - relaxation techniques, muscle relaxants, bite guards
 - patient education = exercises, supplements/herbs, avoid caffeine/alcohol
- **OMT:**
 - mandible, temporal bone (both paired) - common to have *one external rotation and one internal rotation* or asynchronous with respiration, sphenoid bone



- SBS
- Occiput
- Tentorium cerebelli, Dura
- SCM, clavicle, sternum
- Consider muscles of mastication origin and insertion
- Muscles - **lateral pterygoid** muscle inhibition and mandibular muscle energy
- Ligaments - **stylomandibular and sphenomandibular ligament** balancing
- cervical somatic dysfunction
- consider postural imbalance and **leg length discrepancy** (lower extremity)
- treat **sacrum and pelvis** somatic dysfunction (bc of core link)

● Vertigo

- **Temporal Bone** — sometimes one *external* and the other *internal* OR asynchronous rotation
- SBS — compression
- Occiput
- Tentorium Cerebelli (Attached to temporal bone)
- SCM, Clavicle, Sternum
- Sacrum, Pelvis, LE

● Bell's Palsy

- **Temporal Bone** (exit CN 7)
- myofascial restriction in head and neck
- cervical spine
- upper thoracic spine
- pelvis/sacrum
- diaphragms/lymphatic treatments

Intro to BLT/MFR - Wolf

1. Define and differentiate MFR and BLT including treatment principles, procedure, indications/contraindications

Balanced Ligamentous Tension:

Motion of all joints is dependent on

1. Shape of articular surfaces
2. Muscles acting on joint
3. Position of ligaments

Disruption of ligaments maintain somatic dysfunction due to altered distribution of tension. Some ligaments become tense or lax.

Myofascial Release:

Direct or indirect treatment that engages continual palpatory feedback to achieve release of myofascial tissues.

Treatment Principles	
MFR	BLT
Direct or Indirect (or combined) treatment using - Inherent Force - Respiratory Force - Patient Cooperation - Physician-guided force - Springing/Vibration Ex. lymphatics (thoracic inlet/diaphragm release)	Goal is to position joint to balance ligamentous forces. This is achieved by moving joint in direction of the strained ligament using compression/distraction and/or respiratory assistance, until tension within ligaments becomes equal/balanced Ex. OCMM, SI/LS releases

Procedure	
MFR	BLT
1. Positioning into SD barrier (direct) or ease (indirect) 2. Apply activation force 3. Hold and monitor tissues until release 4. Reposition until normal tissue balance is achieved. 5. Reassess	1. Disengage dysfunctional area 2. Exaggerate dysfunctional pattern 3. Balanced tension of ligaments until release 4. Retracing path of injury via ligamentous tension and strain is via respiratory cooperation

Indications/Contraindications			
MFR Indications	BLT Indications	MFR Contraindications	BLT Contraindications
SD	SD	Absolute: - Lack of consent/SD Relative: - Fractures - Open Wound - DVT	Absolute: - Lack of consent/SD Relative: - Fracture/Dislocation - Open Wound - Infection, Hematoma

		- Disseminated Neoplasm - Post-op conditions over site	- Vascular Compromise
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2. Identify key features of functional anatomy that contribute to the maintenance of structural integrity.

- Fascia and connective tissue stabilize and integrate MSK, Nervous, Lymphatic, Visceral, and Vascular tissue
- Fascia: collagenous-based soft tissue in body (tendons, ligaments, bursae, endomysium, perimysium, epimysium)
- Tendons (Allow Movement): **Muscle** → **Bone**; straight fibers
- Ligaments (Stabilize): **Bone** → **Bone**; crimped spring-like configuration

3. Identify clinical implications of the anatomical relationships of myofascial and ligamentous structures

- Pelvis Ligaments
 - Continuity with spine via Anterior Longitudinal Ligament
 - **Iliolumbar Ligaments**: TP of L4/5 to Iliac Crest → Critical in **low back pain**
 - Sacrotuberous Ligament
 - Sacrospinous Ligaments
- Pelvic Floor Muscles
- Psoas: Associated w/ Lumbar TPs & **Low Back Pain**
- Sacrum:
 - Suspended from ligaments so load is equally distributed throughout fascial system
- Sternum
 - Inhalation: Anterior/Superior Movement
 - Exhalation: Posterior/Inferior
 - Divots indicative of **posterior non-physiologic rib**
 - Bumps indicative of **anterior non-physiologic rib**
- Diaphragm
 - Restriction of can cause **rib, lumbar, and xiphoid restriction**
 - Passage of IVC, azygous/homozygous vv., esophagus, aorta
- There is long fascial continuity throughout the body, particularly highlighted by the **visceral fascial layer**: *cranial base (occiput) >> neck >> endothoracic fascia >> endoabdominal fascia >> endopelvic fascia*
- **Sibson's fascia** overlies the thoracic inlet over the copula of the pleura and attaches from the 7th cervical vertebrae to the inner rib 1

4. Define the application of BLT and MFR (including use of both short and long lever for diagnosis and treatment)

- BLT is neither direct or indirect
- BLT/MFR well tolerated by patients with:
 - Fragile Tissues
 - Recent injuries
 - High Pain (Fibromyalgia)
- Short Lever vs. Long Lever
 - Biomechanics
 - Direct Tenderness - try long lever
- Short Lever: palpate spinal structures directly under hand to alter tension of ligaments
 - Palpating short lever in supine: **High yield slide from "BLT/MFR of the Spine, Sacrum, and Pelvis"**



Scan for most dysfunctional segment(s) by reaching behind patient and palpate over transverse processes for increased tension.

Spring anteriorly for restricted rotation

Sidebending can be accomplished by translation

Breath or moving the spinous process inferiorly/superiorly can be used to determine flexion/extension

Long Lever: use extremities or head/neck to alter tension of spinal ligaments

5. Differentiate sprain vs. strain and the different grades

- **Sprain:** Stretching/Tearing of **ligament/joint capsule**
 - Recall that ligaments guide motion, but also restrict motion
- **Strains:** Stretching/Tearing of **muscle/tendons**
- Grading **High yield slide**
 - **1 (Mild):** Some Stretching/Minor Tearing of ligament or muscle
 - **2 (Moderate):** Partially torn but intact
 - **3 (Severe):** Completely torn → joint instability

6. Outline different kinds of non-physiologic ribs and how to diagnose them

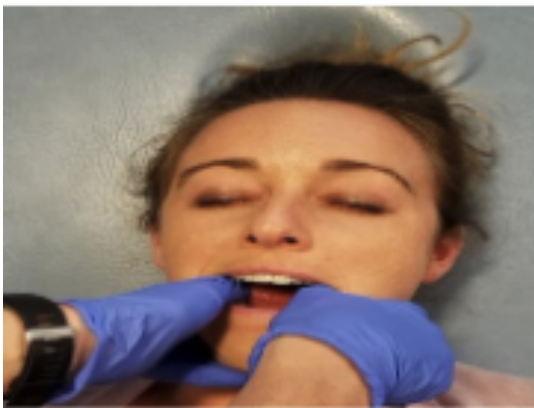
- Non-physiologic ribs are due to trauma, joints not in proper physiological relationship, resulting in ribs that **do not move with respiration**.
 - **Medial Displacement** (Lateral Trauma) — mid-axillary *less prominent* (vs. lateral displacement — mid-axillary *more prominent*)
 - Palpating the sternum:
 - **Bump** at the *chondrocostal articulation*; Divot at the *rib angle* = **Anterior Rib**
 - **Divot** at the *chondrocostal articulation*; Bump at the *rib angle* = **Posterior Rib**
 - **Torsion:** twisting at attachment point, accompanied w/ **Non-neutral/Type 2 c SD**
- Diagnosis
 - Integrated Exam
 - Screen: Asymmetry & Restriction in Breathing Motion
 - Scan/Segmental
 - Springing
 - Inhalation/Exhalation

Week 8 Lab: Osteopathic Approach to TMD - Nuno

- Patients with temporomandibular joint disorders (TMD) present with many different signs and symptoms, including:
 - a specific diagnosis of TMD
 - non-specific jaw pain
 - bruxism
 - headache
 - neck pain
 - dental malocclusion
 - trouble swallowing
 - other areas of somatic dysfunction
- Contraindications:
 - lack of patient consent (absolute)
 - joint instability or dislocation
 - significant pain with manipulation
- Assessment
 - Static; facial symmetry
 - Dynamic; mouth opening/closing
 - Palpation of the TMJ - static and dynamic
 - Cranial vault hold and bitemporal hold
- Techniques
 - **Bilateral Mandibular Direct Muscle Energy (ME)**



- Feathers edge laterally or inferiorly and resist to the opposite direction
- Bilateral Mandibular BLT



- Posterior inferior force until balance is reached
- **Hyoid Muscles Indirect Myofascial Release (MFR)**



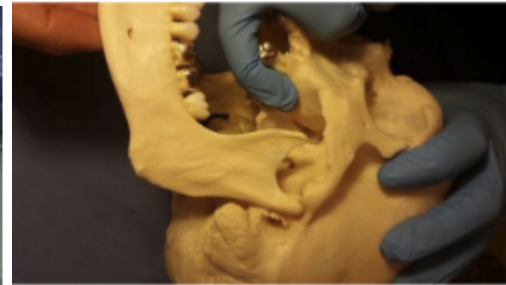
- In ease, stack lateral translation, vertical glide, and rotation >> warmth, softening, neutrality

- Pterygopalatine (Sphenopalatine) Ganglion Inhibition



- With gloved hand, insert the 5th digit on the buccal side of the maxillary teeth. Slide the 5th digit along the buccal surface of the maxillary teeth until the most posterior aspect is reached.
- Direct force superior, anterior and medial

- Lateral pterygoid muscle inhibition



- With gloved hand, insert the 5th digit on the buccal side of the maxillary teeth. Slide the 5th digit along the buccal surface of the maxillary teeth until the most posterior aspect is reached.
- gently apply and hold inhibitory pressure to the muscle until it softens. This area is often exquisitely tender with initial palpation >> tenderness should dissipate with treatment. Warmth or softening indicate a release.